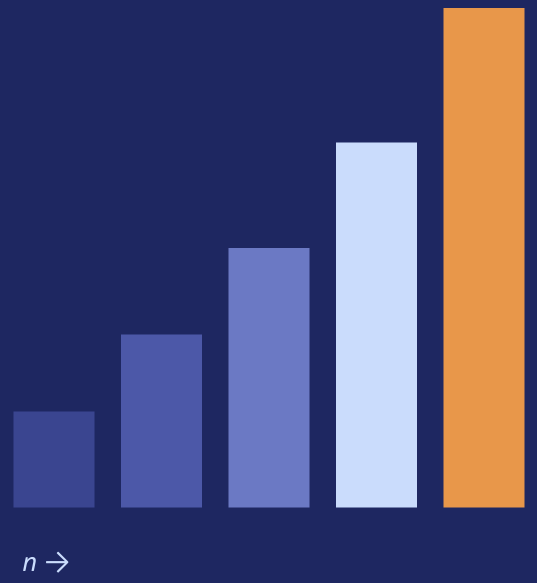
Understanding Growth Functions

A visual comparison of $O(\log n)$, $O(n)$, $O(n \log n)$, $O(n^2)$, and $O(2^n)$



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Presented by:KS THARUNESHWAR

The problem: working with a contact list of 1,000,000 entries — the algorithm you pick decides whether it takes milliseconds or millennia

log n	$O(\log n)$	Binary search a sorted contact list for one name	*~20 comparisons*	Instant (< 1 µs)	**Instant**
n	$O(n)$	Linear scan through every contact to find a match	*1,000,000 checks*	~1 milisecond	**Fast**
n log n	$O(n \log n)$	Sort the entire list alphabetically (merge sort)	*~19,900,000 steps*	~20 miliseconds	**Fast**
n^2	$O(n^2)$	Compare every contact against every other to find duplicates	*1,000,000,000,000 comparisons*	~16.7 minutes	**Slow**
2^n	$O(2^n)$	Brute-force check every possible subset of contacts	*2^1,000,000 combinations*	Longer than the age of the universe	**Infeasible**

KEY INSIGHT

Same 1,000,000 contacts, same hardware — only the algorithm changes. Logarithmic and linear approaches finish instantly; quadratic takes minutes; exponential never finishes at all.

From Growth Rate to Algorithm Choice

Matching complexity class to real algorithms and typical use

log n	$O(\log n)$	Binary search, balanced BST lookup
n	$O(n)$	Linear scan, single-pass array processing
n log n	$O(n \log n)$	Merge sort, quicksort, heap sort
n^2	$O(n^2)$	Bubble/insertion sort, nested-loop comparisons
2^n	$O(2^n)$	Brute-force subsets, unoptimized recursion

Ideal for huge datasets

Scales well, predictable

Best practical ceiling for sorting

Use only for small n

Avoid at scale — redesign or approximate

Conclusion: Efficient algorithm selection means favoring logarithmic, linear, and log-linear growth whenever possible. Quadratic solutions should be reserved for small, bounded inputs, and exponential approaches should be redesigned — through better algorithms, heuristics, or approximation — before they reach production scale.